

The Coherence-Dispersion Sculpting Hypothesis

From Redness and Broccoli to a Testable Multiscale Account of Qualitative Experience

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Abstract

Why does red look different from green, or the taste and crunch of broccoli form one recognizable experience rather than an unstructured collection of neural events? Neuroscience can identify receptors, pathways, population responses, reports, and actions, but correlation at any one scale does not yet explain how the living network constructs a differentiated, multimodal state. This paper develops the Self Aware Networks (SAN) answer as a falsifiable multiscale hypothesis. A qualitative state is proposed to depend on learned, receiver-relative transformations that begin in modality-specific transduction, alter cellular output, recruit recurrent neural arrays, and change an embodied observer-action process. The observer is the distributed process itself; no inner spectator is added.

The Coherence-Dispersion Sculpting Hypothesis (CDSH) supplies one organizing mechanism within this larger construction stack. Neural processing is proposed to depend on locally coherent assemblies, structured phase separation, inhibition, and consequential departures from an ongoing state. SAN calls a typed, receiver-relative departure a phase-wave differential (PWD). The term is introduced only after its ordinary-language function: it names a change in phase, frequency, magnitude, duration, transmission, or network context that a receiving biological system can detect and act upon. Coherence can recruit and bind; structured dispersion can separate, suppress, or preserve boundaries. Neither maximal synchrony nor one privileged frequency is predicted to encode experience.

The broccoli case serves as a qualitative laboratory. Color, shape, texture, pressure, movement, crunch, smell, taste, expectation, and action are measured through different biological routes but learned as a recurrent property bundle. Redness isolates one visual property; umami traces chemical transduction; whisking demonstrates active sensorimotor sampling; ion-channel and action-potential dynamics constrain the molecular-to-synaptic bridge; and neural stimulation and sensory substitution define a bounded engineering question about reconstruction. The paper separates content identity from intensity, persistence, salience, reportability, and motor use. It formalizes local phase organization and a receiver-relative sculpting operator, proposes seven experiments, and states explicit falsification criteria. Seizure, hallucination, psychedelic, lucid-dream, out-of-body, resuscitation, and dying-brain evidence stress-test a non-monotonic dynamic-range prediction: neither maximal coherence nor maximal entropy is expected to optimize every content, access, self-model, report, or action endpoint. Established receptor, oscillation, plasticity, stimulation, and altered-state results are treated as components and constraints, not as proof that the joined SAN account is correct.

Keywords: neural oscillations; phase synchrony; splay state; phase dispersion; spike-timing-dependent plasticity; LTP; LTD; neural tuning; phase-wave differential; neural rendering; predictive processing; Self Aware Networks

Scope and evidence boundary

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1. The Explanatory Target

Suppose a person sees a red vegetable, reaches for it, feels its branching surface, bites it, hears a crunch, and recognizes broccoli. No single receptor receives “broccoli.” Photoreceptors, mechanoreceptors, taste receptor cells, olfactory receptors, proprioceptive systems, and internal bodily signals transform different physical variables. Their outputs travel through partly distinct pathways and are repeatedly modified by attention, expectation, memory, movement, and feedback. Nevertheless, the person experiences a recognizable object with properties and can use that state to guide action.

The scientific target is therefore not merely to decode a stimulus label from brain activity. A decoder can exploit a correlate without identifying what is necessary or sufficient for experience. The target is a causal account that connects at least six questions:

1. How is an external or bodily difference transduced into a biological signal?
2. How does the receiving cell distinguish that signal from its ongoing state?
3. How is the transformed output routed, combined, separated, and recurrently updated across populations?
4. How are content identity, intensity, persistence, salience, and reportability distinguished?
5. How does the distributed state alter action and incorporate the consequences of action?
6. Which perturbations would reproduce, transform, or abolish the relevant qualitative discrimination?

SAN proposes that qualitative experience is constructed through these linked operations. The plain-language core is that every receiver compares what arrives with what is already happening, transforms the difference according to its learned structure and present state, and passes a changed pattern onward. Across many recurrently connected cells and arrays, those differences become a distributed, embodied state that detects, predicts, acts, and updates itself. A **phase-wave differential** is SAN’s name for a typed difference in this process when phase, frequency, magnitude, duration, or coupling is consequential to the receiver. It is not a claim that one oscillation frequency means red or that a neuron literally manipulates a symbolic vector.

Two neurons can repeatedly fire with nearly the same relative timing. They can also fire apart while preserving a stable phase separation. Both organizations can be structured. The first can support coincident arrival, recruitment, or effective coupling. The second can preserve temporal separation, reduce interference, exclude a competing route, or mark a boundary between assemblies. The **Coherence-Dispersion Sculpting Hypothesis** proposes that qualitative neural differentiation depends partly on the changing topology of such local coalitions and boundaries:

Learned, receiver-relative transformations are organized by a reproducible spatial and temporal pattern of local coordination and structured separation; consequential departures alter recruitment, routing, action, and later plasticity.

Communication-through-coherence, phase coding, plasticity, predictive processing, sensory substitution, and stimulation research each constrain parts of this proposal [1-13,26-28]. None alone entails the joined SAN mechanism. The paper therefore labels established components, SAN hypotheses, and present formalizations separately.

2. Historical and Source Genealogy

2.1 Hebbian precursor

Hebb’s 1949 cell-assembly account supplied the classic precursor: repeated co-activity can strengthen effective relations among participating cells [3]. SAN does not originate the claim that co-active cells can become more strongly associated.

2.2 Established plasticity and oscillation precursors

Subsequent work established multiple forms of LTP, LTD, and spike-timing-dependent plasticity. Relative pre- and postsynaptic timing can influence synaptic change, but the learning rule varies substantially by synapse, cell type, state, and protocol [4-6,10]. Oscillatory phase can also affect plasticity. In rat visual-cortex slices, Westpatat and colleagues found LTP when excitatory postsynaptic potentials coincided with peaks of an oscillating membrane potential and LTD near troughs [11]. This is a strong component precedent for phase-sensitive plasticity, not proof of CDSH at network or phenomenal scales.

2.3 The 2017 brain-network protocol question

The April 2017 Neural Lace Podcast asked whether neural communication used connection-and-feedback behavior analogous to TCP, connectionless broadcast behavior analogous to UDP, or a changing combination of both. It proposed **ATP, Alternating Transmission Protocol**, for the hybrid. The acronym was a deliberate double meaning: adenosine triphosphate is cellular energy currency, so the name linked information transmission to biological energy expenditure. The same source asked whether conscious and unconscious traffic used different modes, whether modalities required specialized subprotocols, and whether decoding the protocol could let artificial neurons or chips exchange signals with the brain.

This is an intermediate ancestor, not a backdated PWD or NAPOT formulation. It defines the communication problem and candidate modes. The later PWD account proposes what receiver-relative change may be transmitted, while NAPOT proposes how repeated receive-transform-project cycles can build distributed renderings. The source record therefore supports the genealogy:

protocol question -> feedback / broadcast dichotomy -> alternating hybrid -> modality-specific routing -> phase-wave differential -> neural-array rendering

The local transcript preserves the April 2017 content and original YouTube route, but a public timestamp seal still requires captured platform metadata or contemporaneous archive evidence.

2.4 Public SAN component architecture in 2022

The SAN Git record contains the joined component architecture by June 2022:

1. **a0041z.md**, first committed June 8, 2022, invokes Feynman's path integral, describes phase alignment and cancellation, and immediately asks why two near-phase neurons might reinforce a shared destination. It later returns to LTD while describing mutually timed growth-related signals.
2. **a0142z.md**, committed June 11, 2022, states that coherence supports binding, convergence, and coupling, while decoherence supports distinction, deconvergence, and decoupling. It connects dendritic-spine and receptor structure to microscale LTP-related memory and proposes larger-scale LTD / inhibitory configurations around phasic events.
3. **a0601z.md**, committed June 8, 2022, engages Karl Friston's treatment of value, surprise, motivation, world modeling, and scale-separated behavior. It adds a SAN interpretation of interacting clocks from synapses through circuits, cortical columns, brainwaves, behavior, and culture.
4. **02san.md** defines both unexpected excitation and unexpected inhibition as phase-wave differential events, describes synchronization as preparing communication, and proposes that breaks in synchronization carve a differentiated rendering against the ongoing population state.

The source-faithful 2022 genealogy is therefore:

Hebbian co-activity -> phase-aligned reinforcement and phase-separated cancellation -> LTP / LTD stabilization of timing relations -> coherence for coupling and splay / dispersion for

differentiation -> Friston-related prediction across nested timescales -> SAN rendering, Neural Tuning, and phase-wave differentials.

2.5 Explicit SAN synthesis in 2025

raynote16.md, first committed January 3, 2025 and revised January 13, names Feynman's path integral and Friston's free-energy principle together. It explicitly links splay-phase states, LTP-driven neural synchronization, and wave differences aligning or canceling. The 2025 source is the clearest compact synthesis, but it should not be used to backdate later wording or to erase the component architecture already public in 2022.

2.6 Priority boundary

The potential SAN contribution is not coherence, splay states, LTP, LTD, path integrals, predictive processing, or free energy individually. It is the joined mapping that assigns complementary computational roles to together-firing and persistently separated phase relations, connects those relations bidirectionally to plasticity, and proposes that their distributed organization contributes to qualitative differentiation and observer-action.

2.7 Dated Qualia construction program

The broader SAN Git record shows that the Qualia proposal did not begin with the compact 2025 coherence-dispersion synthesis. Dated 2022 notes separately developed relative tonic and phasic roles, sensory and motor phase fields, distributed detection, persistent re-rendering, artificial qualia, redness and tones, multimodal broccoli properties, smell, mechanosensory receptor dynamics, umami, active whisking, and NAPOT. The present paper joins those source atoms while preserving their distinct dates. The June 12 a0306z.md record already contains an early broccoli-to-BCI route. The explicit redness / tone differential, integrated broccoli property chain, and mature receive-transform-project wording were added on August 23. The explicit SAN interpretation of whisking-related desynchronization was added on November 29. The mechanosensory line must be split again: receptor / threshold reasoning was Git-fixed in June-August 2022, while the current sentence directly tying simultaneous activation to action-potential duration was added January 4, 2024. A separate public SAN Git record, a0395z.md, invokes Oliver Sacks, neurological case studies, split-brain evidence, and the dependence of personality or self-model on brain organization [62]. Sacks is therefore part of the documented SAN intellectual genealogy; his cases do not independently establish CDSH, NAPOT, or the tonic-canvas mechanism.

2.8 Immutable source-atom correction

The source audit dates propositions rather than whole files. That distinction changes four parts of the narrative without erasing their continuity:

1. **Broccoli and redness.** An early June 12 broccoli / BCI proposal survives, while the explicit redness / tone differential, integrated multimodal broccoli chain, and mature NAPOT rendering language are separate August 23 additions.
2. **Distributed observation.** b0100y.md develops the self-detecting recurrent circuit across June 18-19 and explicitly corrects an over-literal picture-frame analogy. The durable claim is that the same distributed process detects and retransmits differences, not that a synapse contains a literal image.
3. **Mechanosensation.** June-August 2022 notes establish receptor coincidence, threshold, ankyrin, and mechanosensory ancestry. The exact action-potential-duration sentence in the current file is a January

2024 addition. Draft 6 therefore preserves the earlier research direction while dating the stronger wording later.

4. **Whisking.** The September 4 file collected whisker and dendritic sources; the explicit claim that desynchronization forms differentiated Qualia relative to tonic activity was added November 29.

The complete atom ledger is reviews/DRAFT4-IMMUTABLE-SOURCE-ATOM-AUDIT-20260720.md. These corrections strengthen priority reasoning by making it falsifiable. They do not convert repository chronology into biological evidence or claim a public timestamp where only Git fixation has been checked.

2.9 Earlier wave-language ancestry

The SAN genealogy also contains relevant precursors to later wave-mind comparisons. A public August 2, 2012 Neo Mind Cycle source describes thoughts as electromagnetic waves produced by coordinated neural firing within an EEG-feedback and mind / body self-change program. Public 2017 Neural Lace material describes neural signals as richer than binary, with amplitude, charge, and tempo-spatial ionic patterns relevant to BCI decoding. These sources predate Gerhab's August 2021 article, but shared vocabulary does not establish mechanism-level equivalence. Exact AM/FM identity, propagation, holographic storage, and medical effects require separate atom-by-atom review.

The distinctive 2022 SAN move is later but more specific: coherence supports binding, convergence, and coupling, while structured decoherence or splay supports distinction, deconvergence, and decoupling. The chronology therefore separates an earlier Micah wave-information narrative, Gerhab's 2021 coherence-first article, and SAN's 2022 complementary-operation account.

2.10 Contribution boundary

Layer	What is inherited	What SAN added in the dated record	What this paper adds
Established biology	Receptor transduction, dendritic integration, action potentials, calcium-dependent release, recurrent circuits, active sensing, plasticity	A proposed joined route from molecular transformation through array-scale rendering and observer-action	A typed construction stack and explicit evidence ceilings
Oscillation science	Phase coding, communication through coherence, state-dependent synchrony, structured oscillator states	Complementary roles for coherence and persistent separation in qualitative differentiation	Operational measures for local coalitions and structured boundaries
2022 SAN Qualia	Redness, tones, broccoli properties, tonic / phasic roles, capture / projection analogies, NAPOT, artificial reconstruction	The distributed observer and receive-transform-project account across modalities	A unified Qualia laboratory and falsifiable reconstruction ladder
2025 SAN synthesis	Feynman / Friston comparison, splay, plasticity, phase alignment and cancellation	Compact coherence-dispersion / plasticity synthesis	Formal separation of analogy, mechanism, and test

No row grants priority over its inherited components. The candidate contribution is the joined architecture and its specific predictions. Draft 6 additionally binds every load-bearing exact-date statement to the proposition-level audit rather than to a file's earliest shell.

2.11 The vocabulary must be earned

The paper follows one editorial and scientific rule: **plain language first, scientific problem second, inherited vocabulary third, SAN terminology fourth, and a discriminating test last**. The rule matters because a coined term should summarize an operation the reader already understands, not substitute a name for an explanation.

The ordinary problem is visible in the broccoli example. Different receptors receive partial properties; each receiver is already in a changing biological state; recurrent populations must combine some signals, keep others distinct, use the result to guide action, and revise the state after new evidence arrives. Established neuroscience describes receptor transduction, dendritic integration, phase-dependent excitability, oscillatory coordination, inhibition, plasticity, active sensing, and recurrent feedback. SAN's proposed operation is narrower and joined: a receiver registers a consequential difference relative to its ongoing and expected state, transforms that difference, and passes a changed pattern into a recurrent observer-action process. Only after that operation has been stated does this paper use **phase-wave differential** as its compact name.

PWD is therefore not another name for phase coding or a traveling wave. Phase coding assigns information to timing relative to a reference cycle. Traveling-wave analysis describes spatial propagation of phase or activity. A PWD is a typed, receiver-relative event whose significance depends on the arriving signal, the receiver's current state, the relation expected on that route, and the downstream change it produces. A PWD may involve phase, frequency, magnitude, duration, inhibition, omission, propagation, or coupling; a traveling wave or phase code can carry or shape such an event without exhausting its definition.

The corresponding test is incremental. If conventional phase, power, rate, connectivity, and behavior variables explain an intervention's effects as well as the receiver-relative PWD variables, the new term has not earned additional explanatory value in that experiment. If the receiver-relative model predicts selective changes in discrimination, reconstruction, action, or plasticity beyond those matched baselines, it has.

2.12 Load-bearing source custody

The source chain is governed at proposition level. A publication date, a local file's internal date, a Git fixation date, and the date when stronger wording was added are recorded as different facts. The paper does not use a later sentence to backdate a weaker ancestor, and it does not erase an earlier component because the mature term arrived later.

For the external literature, the most load-bearing claims are tied to exact sections, figures, or pages in reviews/DRAFT6-LOAD-BEARING-SOURCE-LOCATOR-AUDIT-20260727.md. The anchors include Hebb's cell-assembly and phase-sequence account [3]; oscillation-phase-dependent LTP / LTD in rat visual cortex [11]; the time-dependent results of Anisimova et al. [12]; synchronization / desynchronization changes in STDP models [13,25]; mathematical generalized splay states [15]; the measurement limits of gamma coherence [17]; phase-sensitive hippocampal plasticity [21,22]; human theta-gamma coupling and memory [23]; and a computational bridge between theta phase and STDP [24]. These sources establish components or constraints. The joined CDSH mechanism and its phenomenal interpretation remain the paper's hypotheses.

3. The Qualia Laboratory

3.1 Redness: one property, many transformations

Redness is a useful first case because the physical stimulus, receptor responses, cortical population activity, discrimination, report, and action can be manipulated separately. Distributed activity in human visual cortex can support decoding and reconstruction of stimulus color [29], but successful decoding does not show that

a decoded pattern is sufficient for the experience or that color is stored in a single place. When retinal stimulation was held fixed while perceived color changed, responses in V4 and VO1 tracked the reported percept more closely than responses in V1 and V2, which more strongly reflected the stimulus [40]. This dissociation constrains the construction route without establishing that any one area is sufficient for redness. SAN instead treats redness as a learned relation within the current recurrent state. The relevant state includes the stimulus-driven transformation, ongoing adaptation, neighboring colors, attention, memory, bodily orientation, and downstream consequences.

The proposal is therefore relational and receiver-specific. A signal has no fixed qualitative meaning solely because it has a particular frequency. It contributes to redness when the signal is transformed through the relevant learned visual pathways and participates in a reproducible distributed state that discriminates, predicts, and guides behavior. This yields a stronger experimental requirement than correlation: interventions should selectively alter red-related discrimination or experience while matching rate, power, arousal, and task demand.

3.2 Broccoli: a multimodal property bundle

Broccoli exposes what a single-property example leaves hidden. Its visual color and branching shape, resistance under the fingers, movement toward the mouth, crunch, smell, savory taste, and learned expectations originate in different receptor populations and pathways. Their unity is not present at the receptor surface. It is learned through repeated temporal and spatial co-occurrence, recurrent routing, bodily sampling, and consequences.

SAN's 2022 worked example proposes the following bounded chain:

external or bodily property -> receptor-specific transduction -> dendritic and cellular transformation -> patterned output -> local and distributed array recruitment -> recurrent multisensory rendering -> discrimination and action -> reafferent update

The word *rendering* here names the continued biological transformation and projection of a state into states that other cells and arrays can detect. It does not name a picture placed before a hidden viewer. The rendered broccoli state can be incomplete, expectation-laden, and task-dependent. Its visual components may dominate at a distance; touch, smell, taste, and motor predictions are added or revised as the organism samples the object.

3.3 Umami: chemical transduction is already computation

Umami supplies a molecularly specific case. T1R1 and T1R3 form a heteromeric G-protein-coupled receptor responsive to L-amino acids, with receptor sequence and ligand context affecting selectivity [30]. T1R3 knockout mice nevertheless retained diminished behavioral and nerve responses to sugars and umami compounds, indicating T1R3-independent receptors or pathways [41]. Umami is therefore a distributed transduction problem rather than a one-receptor code. This demonstrates that the first biological transformation is neither a neutral relay nor a universal scalar code. Receptor composition, intracellular signaling, membrane conductances, and the state of the taste cell shape its downstream output.

SAN uses umami to ask how chemically selective reception becomes a tempo-spatial population pattern that can be integrated with texture, smell, expectation, and action. The established receptor mechanism does not prove a PWD account. It constrains one layer of it: any SAN model must respect the actual receptor populations, transduction pathways, species differences, and temporal response properties rather than replace them with one generic oscillation.

3.4 Whiskers: experience is actively sampled

Whisking shows why qualitative construction cannot be modeled as passive input recording. In awake mice, quiet and active whisking states exhibit different membrane-potential dynamics; active whisking can reduce local membrane-potential correlation, and object contact produces behavior-dependent responses [31,32]. In a head-fixed localization task, a model combining touch count with whisking midpoint explained behavior better than several more elaborate timing and geometry models [42]. That result is a serious non-phase null: active sampling can organize discrimination through low-dimensional sensorimotor variables without requiring the CDSH operator. The sensory evidence therefore depends on movement and prestimulus state.

SAN interprets this as a demanding test of tonic context and phasic difference. Movement changes what is sampled, when it arrives, and which populations are synchronized or desynchronized. The resulting tactile distinction is incorporated into an updated sensorimotor state and can alter the next movement. Structured desynchronization is not automatically loss of information; it may increase differentiation or signal-to-noise in the task-relevant route. That claim must be tested against rate, movement, arousal, and common-input explanations.

3.5 Potassium, action-potential waveform, calcium, and release

The historical SAN notes connect potassium-channel state to action-potential duration, calcium entry, and vesicle release. The defensible modern statement is conditional and compartment-specific. Membrane voltage and action-potential waveform arise from interacting conductances, gating kinetics, ionic gradients, morphology, temperature, and recent activity. At presynaptic terminals, action-potential shape influences voltage-gated calcium-channel activation; calcium entry controls transmitter release through highly nonlinear and synapse-specific coupling [33-36]. Release remains probabilistic, and multivesicular release is not a universal count encoded by bulk intracellular potassium.

This refined chain preserves SAN's important multiscale question: can molecular and cellular differences change the timing, magnitude, duration, and reliability of population signals in ways that contribute to qualitative content or vividness? It removes the unsupported shortcut that one potassium quantity directly specifies one vesicle count. The experimental variables are channel identity and localization, waveform, calcium current, release statistics, postsynaptic effect, and later network participation.

3.6 The reconstruction question

Sensory substitution establishes that patterned information from an artificial receptor can become behaviorally useful when delivered through another sensory interface [37,38]. Intracortical and thalamic stimulation can evoke localized tactile percepts and improve prosthetic control [26-28]. These findings make neural read / write research plausible without demonstrating that a complete multimodal object experience has been captured or transferred.

The broccoli proposal should therefore be stated as an engineering ladder, not an accomplished result:

1. identify repeatable neural features associated with naturally sampled broccoli properties;
2. test held-out decoding across context, action, and time;
3. perturb one property while preserving the others;
4. reproduce a localized, discriminable component through stimulation;
5. test whether multiple reproduced components bind into a learned object state;
6. test behavioral use, recurrence, reversibility, and subject-specific calibration; and
7. only then ask whether cross-person translation preserves any qualitative equivalence.

Different brains need not store broccoli in identical cortical locations or use identical detailed codes. A successful interface may require learning a translation between each person’s receptor-to-action organization rather than copying a universal spatial template.

4. Definitions

4.1 Relative phase

For oscillator-like neural signals with instantaneous phases $\theta_i(t)$ and $\theta_j(t)$, define the relative phase

$$\Delta\theta_{ij}(t) = \text{wrap}(\theta_i(t) - \theta_j(t)). \quad (1)$$

The relevant phase may be derived from membrane potential, local field potential, population rate, spike phase relative to a declared rhythm, or another measured state variable. These are not interchangeable measurements.

4.2 Pairwise phase consistency

Across K observations or time windows, a conventional pairwise phase-consistency estimate is

$$C_{ij} = \left| \frac{1}{K} \sum_{k=1}^K e^{i\Delta\theta_{ij,k}} \right|, \quad 0 \leq C_{ij} \leq 1. \quad (2)$$

High C_{ij} indicates a stable relative phase, not necessarily zero-lag synchrony. Two cells can be consistently phase-offset and still have high phase consistency.

4.3 Population order parameter

For N oscillatory units,

$$R(t)e^{i\Psi(t)} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)}. \quad (3)$$

Here $R \approx 1$ indicates concentrated phases, while $R \approx 0$ can indicate random incoherence, balanced antiphase clusters, or an ordered splay state. Therefore R alone cannot distinguish structured dispersion from noise.

4.4 Structured phase dispersion

Let $p(\theta)$ be the phase distribution and $U(\theta) = 1/(2\pi)$ the uniform distribution. A splay state can have phases distributed approximately around the cycle with stable ordering. CDSH therefore requires at least two measurements:

1. **dispersion**, describing the spread of phases; and
2. **temporal structure**, describing whether relative offsets persist or recur.

One bounded measure is

$$D_{\text{struct}} = H[p(\theta)] Q_{\text{offset}}, \quad (4)$$

where $H[p]$ is normalized circular entropy and Q_{offset} measures the reliability of pairwise phase offsets across time. Random phase noise may have high entropy but low offset reliability. A stable splay state can have both high dispersion and high offset reliability.

4.5 Local assemblies and boundaries

Partition a measured population into candidate assemblies $A_1, \dots, A_M$. Define

$$C_{\text{within}} = \frac{1}{M} \sum_{m=1}^M \frac{1}{|E_m|} \sum_{(i,j) \in E_m} C_{ij}, \quad (5)$$

and a between-assembly separation term

$$B_{\text{between}} = \frac{1}{|E_B|} \sum_{(i,j) \in E_B} g(\Delta\theta_{ij}, Q_{ij}, I_{ij}), \quad (6)$$

where Q_{ij} is offset reliability, I_{ij} is measured inhibitory or suppressive influence, and g is preregistered for the experiment. CDSH predicts informative states with high within-assembly coordination and nonrandom between-assembly separation, not maximal global R .

4.6 Phase-wave differential event

For receiver r , define an expected tonic state $\hat{\mathbf{s}}_r(t)$ and observed incoming or local state $\mathbf{s}_r(t)$. A phase-wave differential event is

$$\mathbf{d}_r(t) = \mathbf{s}_r(t) - \hat{\mathbf{s}}_r(t), \quad (7)$$

with a typed state vector such as

$$\mathbf{s}_r = [\phi, f, A, \tau, q, \rho, \mathbf{x}, \mathbf{c}]^T, \quad (8)$$

where phase, frequency, amplitude, duration, transmitted quantity, variability, position, and network context are declared separately. Equation (7) is a model definition, not a claim that neurons symbolically subtract vectors.

5. The Multiscale Construction Stack

The worked examples motivate a six-level stack. The levels are analytical boundaries, not a claim that biological processing proceeds in one feedforward line.

Level	Biological operation	Established constraint	SAN hypothesis	Measurement burden
Transduction	Physical or chemical energy changes receptor state	Modality-specific receptors and cascades transform different variables	Receptor dynamics initiate typed tempo-spatial differences	Receptor current, conductance, ligand or stimulus, adaptation
Cellular comparison	Dendrites and membrane state integrate convergent input	Integration depends on morphology, location, timing, inhibition, and state	The receiver detects a PWD relative to its ongoing context	Compartment voltage, spikes, phase, dendritic calcium, inhibition
Output transformation	Axon and terminal convert cellular state into patterned influence	Waveform and presynaptic calcium constrain probabilistic release	Duration, magnitude, timing, and target distribution carry more structure than a binary spike label	Terminal waveform, calcium current, release statistics, target response
Array organization	Local populations recruit, separate, and route activity	Coherence, inhibition, and phase relations affect effective communication	Coherence-dispersion topology sculpts content and boundaries	Unit / LFP phase, directed influence, assembly identity, perturbation
Recurrent rendering	Distributed pathways repeatedly transform and re-enter state	Perception depends on recurrent and state-dependent processing	Repeated receive-transform-project cycles construct differentiated experience	Cross-area timing, recurrence, content decoding, causal disruption
Observer-action	Body and movement change the next input and internal state	Active sensing and reafference alter perception	The distributed loop itself performs observation; action is constitutive, not appended	Behavior, proprioception, autonomic state, reafference, report

The table prevents two common collapses. First, evidence at one level cannot automatically validate the levels above it. A receptor mechanism does not prove a phenomenal theory, and a content decoder does not prove sufficiency. Second, the levels cannot be treated as independent modules. Whisking changes transduction and cortical state; expectation changes dendritic and population responses; action changes the next sensory evidence.

5.1 Capture and projection are functional roles

SAN's camera / pixel and microphone / speaker analogies describe paired functions. Receptors and dendrites capture or register differences; axons and terminals project transformed influence. A neuron is not literally a camera, screen, microphone, or loudspeaker, and no cell contains a miniature image or sound. The explanatory point is that the same distributed tissue participates in reception and transmission. At the network scale, ongoing population activity is altered by each consequential output and becomes the context against which subsequent differences are registered.

5.2 Tonic and phasic are relative roles

The tonic state is the ongoing receiver-relative context, including expectations and active assemblies. A phasic event is a consequential departure from that context. These roles are not permanently assigned to delta, theta, alpha, beta, or gamma bands. A slow event can be phasic relative to a faster local baseline, and high-frequency activity can be tonic within a sustained local process. This avoids mapping named frequency bands directly onto fixed meanings.

5.3 Content is not intensity

For a qualitative state (q), distinguish at least five observables:

$$\mathbf{q}(t) = [I_{\text{id}}, I_{\text{int}}, T_{\text{persist}}, S_{\text{sal}}, P_{\text{report}}]^T. \quad (9)$$

Here identity denotes which discrimination is made, intensity denotes graded strength, persistence denotes duration or recurrence, salience denotes competition for processing and action, and reportability denotes access to a particular behavioral report. These variables can covary but are not interchangeable. A brighter red can preserve color identity; an unreported state can still guide action; a persistent afterimage can outlast the stimulus; a salient distractor need not be vivid.

6. The Coherence-Dispersion Sculpting Operator

Let $\mathcal{N}_r(t)$ denote the receiver's current effective neighborhood, including excitatory, inhibitory, and modulatory influences. Define the proposed sculpting state

$$\mathcal{S}_r(t) = F(C_{\text{within}}, B_{\text{between}}, \mathbf{d}_r, \mathcal{N}_r, \mathbf{b}(t)), \quad (10)$$

where $\mathbf{b}(t)$ includes bodily, behavioral, and task state. The next effective routing state is

$$\mathcal{N}_r(t + \Delta t) = G(\mathcal{N}_r(t), \mathcal{S}_r(t), \mathbf{a}(t), \mathbf{y}(t)), \quad (11)$$

where $\mathbf{a}(t)$ is action and $\mathbf{y}(t)$ is sensory or bodily consequence. This makes observer-action explicit: a neural state is not complete when represented internally; its motor and refferent consequences constrain the next cycle.

CDSH assigns four complementary operations:

1. **Recruitment:** temporally compatible inputs increase the probability that a receiver or assembly participates.
2. **Binding:** stable relative timing supports coordinated influence among selected participants.
3. **Separation:** structured offsets and inhibition reduce interference or maintain boundaries between assemblies.
4. **Reconfiguration:** a PWD event changes which relations will be recruited, suppressed, or learned next.

7. Plasticity: Fast Organization and Slow Stabilization

7.1 No universal coherence-to-LTP rule

For a synapse with weight w_{ij} , a generic timing-dependent update can be written

$$\Delta w_{ij} = \int F_{ij}(\Delta t; \mathbf{z}) p(\Delta t | \mathcal{S}, \mathbf{z}) d\Delta t, \quad (12)$$

where F_{ij} is the pathway-specific plasticity kernel and $\mathbf{z}$ includes postsynaptic voltage, dendritic location, neuromodulation, prior activity, and other state variables. Coherence or dispersion changes the distribution of relative timings; it does not determine the sign of F_{ij} universally.

7.2 Bidirectional loop

The proposed slow update is

$$\mathbf{W}_{t+1} = \mathbf{W}_t + \eta \mathcal{P} \left(\{\mathcal{S}(t')\}_{t-T}^t, \mathbf{z} \right), \quad (13)$$

and the next fast organization depends on the changed structure:

$$\mathcal{S}_{t+1} = \mathcal{F}(\mathbf{W}_{t+1}, \mathbf{u}_{t+1}, \mathbf{b}_{t+1}). \quad (14)$$

Thus fast phase organization biases plasticity, while plasticity reshapes future phase organization. This is the core coherence-dispersion / plasticity loop.

7.3 Existing support and limits

Wespatat et al. provide direct slice evidence that synaptic modification can depend on input phase relative to membrane oscillations [11]. Anisimova et al. found short-term timing-dependent LTP and LTD but stable potentiation for both causal and anticausal paired activity after three days, not asynchronous patterns [12]. Additional hippocampal, human-memory, and modeling studies show that phase, cross-frequency timing, and the chosen plasticity rule can alter potentiation or synchronization, but the relevant sign and persistence remain preparation-specific [21-25]. These findings support context-sensitive timing effects and warn against a simple binary rule. Computational studies show that STDP can itself alter the duration and recurrence of synchronized and desynchronized episodes [13,25]. CDSH goes beyond these results by predicting that a structured population pattern of local coherence and between-assembly dispersion helps organize both present content and later plasticity.

7.4 Three layers that must not be collapsed

The genealogy and the empirical claim are clearer when three layers remain distinct.

1. **Hebbian precursor.** Hebb proposed that repeated co-activity can stabilize cell assemblies and that ordered assembly activity can form phase sequences [3]. This is the historical backbone for together-active cells becoming more likely to participate together. It does not by itself specify oscillatory coherence, structured splay, a receiver-relative PWD, or the construction of qualitative content.
2. **Measured phase-sensitive plasticity.** Specific experiments and models show that relative timing and oscillatory phase can change the sign or magnitude of plasticity, and that plasticity can change later synchrony [11-13,21-25]. The direction depends on preparation, delay, phase, duration, cell type, state, and plasticity rule. These results justify a bidirectional timing-plasticity bridge, not a universal equation of coherence with LTP or dispersion with LTD.
3. **SAN's added proposition.** SAN proposes that fast organization has two complementary structured forms: locally compatible timing can recruit and bind a candidate coalition, while persistent phase separation, inhibition, and interference can preserve distinctions or exclude competing routes. The history of those relations can bias synaptic change; the changed structure then alters which coalitions and boundaries are available in the future. CDSH further proposes that this evolving coalition-boundary topology helps differentiate qualitative states.

This third layer can fail independently of the first two. The SAN proposition would be weakened if matched perturbations of structured coordination and structured separation did not improve prediction of later plasticity, content discrimination, or observer-action consequences beyond established timing, rate,

power, connectivity, and behavioral models. It would also be weakened if apparent splay effects vanished when phase dispersion was matched by unstructured noise. Conversely, a result showing phase-dependent LTP or LTD alone would not establish the complete CDSH account.

8. From Neural Coordination to Qualitative Differentiation

8.1 No inner viewer

CDSH does not place perception inside one neuron, cortical area, field center, or display. The observer is the distributed recurrent biological process. Cellular receivers and transmitters, ongoing population activity, memory-dependent connectivity, bodily signals, and action consequences collectively produce discrimination and response. In SAN's functional analogy, dendrites and receptors are internal capture surfaces and axon terminals are internal projection surfaces. The analogy earns explanatory value only when embedded in the tonic field: each projected change modifies the population state that other cells receive. The population's changed state is the observation, not an image waiting for a second agent to inspect it.

An operational observer criterion should therefore require a closed causal role. A candidate state must (i) register a difference, (ii) preserve or transform it in a way that changes later discrimination, (iii) distribute that change to other parts of the system, and (iv) affect prediction, action, learning, or reafferent processing. A passive correlate that can be decoded externally but has no demonstrated role in the system's own next state does not satisfy this criterion.

8.2 Tonic canvas and phasic ink

At any declared scale, a tonic canvas is the receiver's ongoing context. Phasic ink is a consequential departure from that context. The departure can be:

- an extra spike or burst;
- inhibited or omitted activity;
- a changed phase relation;
- a changed duration or transmitted quantity;
- recruitment into an assembly; or
- exclusion from an assembly.

The canvas is therefore not homogeneous synchrony. It is an organized background of ongoing relations against which changes become detectable. Low information in this relative sense does not mean physically inactive or semantically empty. A stable or predictable tonic state can have lower surprise under a declared model, while a PWD can carry greater update information because it is less expected. This Shannon-style distinction concerns probability relative to a model; it does not make tonic activity universally low entropy or PWDs universally high entropy.

8.3 Qualia claim

The strongest SAN claim is:

The differentiated qualitative structure of a moment depends partly on the geometry of locally coupled assemblies and the structured phase boundaries separating them.

This claim is not established by showing that synchrony correlates with perception. It requires content-specific prediction. Two experiences with similar power and mean synchrony should differ in reproducible

coherence-dispersion topology, and perturbing that topology should alter discrimination or report in a predicted direction while controlling for rate, power, arousal, and task demand.

The claim also does not equate a qualitative state with its verbal report. A state may alter forced choice, generalization, autonomic response, motor preparation, or later memory without immediate verbal access. Conversely, a report may be generated from expectation or task strategy. In a no-report masking study, the P3b was present when participants reported but absent when they did not, despite perceptual visibility, demonstrating that a candidate neural signature can primarily reflect postperceptual reporting [45]. Experiments should therefore use converging readouts and distinguish neural content identity from report preparation.

8.4 Operational observer-action criterion

The distributed-observer claim becomes testable only when the candidate state has a role in the organism’s own causal cycle. Let Z be a preregistered neural state representation. Define four held-out quantities: $D(Z)$, content discrimination beyond nuisance variables; $C(Z)$, selective causal sensitivity to intervention; $R(Z)$, prediction of action consequence or reafferent update; and $G(Z)$, generalization beyond the report task. A domain-specific observer-action gate is

$$\mathcal{O}(Z) = \mathbf{1} [D(Z) > \delta_D \wedge C(Z) > \delta_C \wedge R(Z) > \delta_R \wedge G(Z) > \delta_G]. \quad (15)$$

The thresholds must be fixed before analysis and compared with rate-, power-, movement-, arousal-, and report-matched null models. Passing $\mathcal{O}(Z)$ would show that the state participates in a closed observer-action process in the tested domain. It would not by itself detect consciousness, prove phenomenal sufficiency, or settle whether another system has the same experience.

9. Relation to Existing Frameworks

9.1 Communication through coherence

Communication through coherence proposes that aligned excitability windows make interareal influence effective [1,2,17,19,20]. CDSH accepts effective alignment as one component but asks what organized nonalignment contributes. It predicts that structured dispersion is not merely transmission failure; it can preserve boundaries, suppress interference, and prepare alternative routes.

9.2 Phase coding

Phase coding asks whether event timing relative to an oscillatory cycle carries information [7,16,23]. CDSH asks how relative phase organization among receivers and assemblies changes recruitment, suppression, routing, and plasticity. Phase coding can exist without the full sculpting hypothesis.

9.3 Winner-take-all and lateral inhibition

Competitive inhibition selects among candidate responses. CDSH treats inhibition as one mechanism capable of creating and maintaining assembly boundaries. It does not claim that every splay state is produced by winner-take-all circuitry or that one canonical inhibitory circuit is universal.

9.4 Predictive processing and free energy

Friston’s free-energy principle and predictive-coding microcircuit proposals supply formal accounts of perception and action under generative models [8,9]. CDSH proposes a candidate physical implementation layer: receiver-relative differences alter phase organization and routing, and action returns consequences that constrain the next state. This bridge is a SAN hypothesis. Free-energy minimization does not entail phase-wave differentials or coherence-dispersion sculpting.

9.5 Feynman’s path integral

In the path integral, possible histories contribute complex amplitudes whose phases reinforce or cancel [14]. SAN uses this as a conceptual and mathematical analogy for reinforcement and cancellation among phase-organized neural routes. The paper does **not** propose that ordinary neural learning requires quantum path-integral computation. Demonstrating a formal analogy is not evidence of a shared physical mechanism.

9.6 Splay states

In coupled-oscillator theory, a splay state distributes oscillator phases around the cycle and can yield a low global order parameter despite precise internal organization [15]. CDSH imports this distinction between structured dispersion and random incoherence. Biological applicability requires identifying the measured oscillator, coupling, persistence, and circuit mechanism rather than assigning the mathematical label from a visually similar trace.

9.7 Coherence, decoherence, and interference

The vocabulary of coherence and interference appears in other wave accounts of mind. Gerhab, for example, proposes that coherent neural waves amplify signals and form holographic memories, while incoherent waves are weaker [39]. That article is useful as a comparison target, but its electromagnetic propagation, stimulated-emission, holographic-memory, and health claims are not adopted here and do not provide validation for SAN.

CDSH instead separates three operations:

1. **Coherence or stable phase relation:** selected events maintain a reproducible timing relation and can reinforce effective communication.
2. **Structured decoherence or dispersion:** events remain separated in a reproducible way and can preserve boundaries, alternatives, or reduced interference.
3. **Interference or interaction:** coherent and phase-separated activity jointly changes a receiver’s membrane, phase, recruitment, or output. Constructive and suppressive consequences produce the changing differential pattern that SAN calls candidate mental ink. They are measured biologically rather than inferred from an optical hologram analogy.

In this paper, *decoherence* does not mean loss of quantum coherence. It denotes a declared neural phase relation becoming less concentrated or changing its prior organization. Random dispersion, ordered splay, inhibition, intermittent synchrony, and content-specific phase reconfiguration must still be distinguished empirically [18]. SAN’s candidate contribution is that all three regimes can carry computational structure: coherence can be “on-together,” structured splay can be “apart-together,” and their interaction can change the rendered content. Coherence is not automatically healthy or informative, and dispersion is not automatically failure.

9.8 Altered states stress-test the dynamic range

The strongest test of the coherence-dispersion hypothesis is not ordinary perception, where arousal, sensory evidence, self-location, source attribution, and action usually agree. Seizures, hallucinations, psychedelic states, lucid dreams, out-of-body experiences, and reports around cardiac arrest can dissociate these variables. Their differences prevent a single “more coherence” or “more entropy” explanation.

Epilepsy supplies the clearest warning against a monotonic coherence account. Synchronization and desynchronization can both occur during seizures, and their timing, spatial scale, cell population, and measurement method matter [50]. During focal-to-bilateral tonic-clonic seizures, loss of consciousness has coincided with whole-brain high-gamma increases and especially synchronous deep-source activity, whereas focal impaired-awareness seizures can show a different suppression-dominant pattern [51]. Excessive synchrony is therefore not a stronger version of ordinary binding. It can coexist with the loss of the flexible differentiation and access required for organized experience and behavior.

Seizures can also generate positive visual content rather than simply erase it. Occipital seizures can produce brief stereotyped multicolored circular hallucinations [55]. Under visual deafferentation, Charles Bonnet hallucinations can remain vivid and organized, with category-related activation in visual association cortex [49]. A newer multimodal comparison of nineteen people with Charles Bonnet syndrome and eighteen vision-matched controls found reduced visually evoked BOLD responses, reduced alpha reactivity, greater occipital theta power, and converging evidence of altered visual-cortical excitability [65]. A separate six-participant event-linked EEG study found lower occipital alpha power at hallucination onset than at offset [66]. These observations constrain any tonic-canvas account: current retinal input is not required for every visual content, but internally generated activity must still acquire structure, persistence, and a source interpretation. Conditioning can also induce perception-like reports when learned priors outweigh ambiguous sensory evidence [60], while integrative reviews conclude that hallucinations arise through multiple interacting routes rather than one universal lesion, rhythm, or inference failure [61]. The discriminating question is therefore which typed neural relations predict content and source attribution beyond expectation, sensory reliability, medication, arousal, and report strategy. Baseline susceptibility and event-linked transition are distinct variables.

Psychedelic findings challenge the opposite monotonic claim. Several MEG signal-diversity measures increased under ketamine, LSD, and psilocybin in one analysis [57]. In a densely sampled crossover study, psilocybin produced widespread functional-connectivity reorganization and spatial desynchronization; the changes tracked subjective intensity and were reduced during an externally constraining perceptual task [67]. Yet a 2026 independent psilocybin study found that eight of fourteen fMRI entropy metrics did not show consistent effects, Lempel-Ziv effects were inconsistent, and the metrics had limited intercorrelation [54]. “Brain entropy” is not one scalar and cannot be identified with richness, consciousness, decoherence, or meaning without naming the estimator, signal, scale, and endpoint. The task effect is especially relevant to SAN because it tests whether external constraint changes the departure from an internally maintained state.

Lucid dreaming provides another useful dissociation. An early six-participant study reported spectral and coherence differences, including a gamma peak, during signaled lucid REM episodes [58]. A later pooled multilaboratory analysis found minimal sensor-level differences, broad low-alpha-to-gamma power reductions relative to wake, source-level beta reductions, and increased alpha connectivity relative to nonlucid REM [47]. Meanwhile, limited two-way communication has been demonstrated during verified REM sleep in correctly answered trials from six of thirty-six participants across four laboratories [52]. Lucidity is therefore not one band. It is a conjunction of an internally generated scene, partial restoration of metacognitive or source access, and enough preserved action routing to emit a trained signal.

Out-of-body experiences separate scene content from bodily self-location. Electrical stimulation near the

right angular gyrus repeatedly altered body and self-location experience in one epilepsy patient [46]. Visual-tactile conflicts in healthy participants can also shift self-location, identification, and threat-related responses [48,53]. These results do not establish literal departure from the body or a single OBE center. They show that viewpoint, ownership, and self-location are constructed relations that can be perturbed separately.

Reports around cardiac arrest require an especially strict timing boundary. AWARE-II recorded physiology during resuscitation and recovered several categories of remembered experience from survivors, while also documenting the possibility of organized EEG activity during portions of prolonged cardiopulmonary resuscitation [56]. In a different retrospective dataset, two of four comatose dying patients showed selected gamma and connectivity increases after withdrawal of support, but none survived to report experience [59]. Neither report content nor a metaphysical conclusion can be inferred from gamma alone. Cardiac arrest, ongoing chest compression, return of circulation, coma, medication, delirium, dream-like experience, later memory, and irreversible brain death are distinct states.

These contrasts motivate a receiver-relative dynamic-range hypothesis. Let $\mathbf{s}_r(t)$ be the measured state available to receiver r , and let $\hat{\mathbf{s}}_r(t \mid h_t, z_t)$ be a tonic reference estimated without future leakage from history h_t and declared state covariates z_t . The typed departure is

$$\delta_r(t) = \mathbf{s}_r(t) - \hat{\mathbf{s}}_r(t \mid h_t, z_t). \quad (16)$$

The departure can include declared phase, timing, amplitude, duration, burst, spatial, coupling, and receiver-consequence components. Unlike units remain separate unless a normalization and uncertainty model is declared.

Let C_W measure coordination within a candidate assembly, D_S measure reliable structured separation among assemblies, D_R measure phase-scrambled or otherwise unexplained dispersion, G measure current sensory or mnemonic constraint, and B denote the baseline state. The hypothesis is not that content or access rises monotonically with global coherence or global entropy:

$$P(Y_{t+1} \mid \mathbf{s}_r, h_t, z_t) \neq f(C_{\text{global}}) \quad \text{and} \quad P(Y_{t+1} \mid \mathbf{s}_r, h_t, z_t) \neq g(H_{\text{global}}) \quad (17)$$

for universal monotone functions f or g . The positive prediction is model-comparative. A structured receiver-relative model must improve held-out prediction of a prespecified content, source, self-location, report, or action endpoint:

$$\Delta_{\text{CDSH}} = \mathcal{L}_{\text{test}}(M_{C_W, D_S, D_R, G, B}) - \max_{m \in \mathcal{M}_0} \mathcal{L}_{\text{test}}(m), \quad (18)$$

where $\mathcal{M}_0$ contains equal-information, approximately equal-capacity rate, power, global-coherence, entropy, state, and recurrent-history baselines. Support requires preregistered positive incremental value, calibrated uncertainty, replication, and - where ethically possible - selective perturbation with component-matched rescue.

This model treats saturation and randomization as different failures. Excessive global coordination can reduce separability, flexible routing, or repertoire. Unstructured dispersion can reduce reliability and causal access. A productive regime contains an ongoing context plus structured departures: enough local coordination to make an assembly effective, enough structured separation to preserve alternatives and boundaries, and enough external or mnemonic constraint to stabilize source and meaning.

9.8.1 A bounded phenomenological case: the pastel room

Ken Wilber reported multiple generalized seizures, several days of coma, a dim awareness of the hospital room, and an altered or absent ordinary autobiographical subject [63]. In a separate first-person health account, he described repeatedly seeming to awaken from a dream of a blue-and-pink pastel room only to encounter the same apparent room [64]. He interpreted the remaining awareness as “Big Mind.” The report is relevant because it separates visual scene, source attribution, autobiographical self, awareness, and later memory. It does not identify the neural state that produced each remembered episode.

The scientific use of the case is therefore limited but real. It motivates synchronized measurement of scene content, source judgment, self-location, autobiographical access, physiology, intervention timing, and later report. It does not license a retrospective diagnosis, prove experience during the tonic-clonic events themselves, identify “Big Mind” with a tonic oscillation, or establish awareness during an electrically silent interval.

“Big Mind” is not a biomedical variable. Its closest operational research neighbors are nondual awareness, content-free awareness, reduced self-world distinction, and altered internal-versus-external processing. A single-expert EEG-fMRI case associated a reported content-free state with reduced alpha power, increased theta power, increased dorsal-attention-network connectivity, and decreased posterior default-mode-network connectivity [68]. A larger 2026 EEG study operationalized nonduality as a reduced difference between internal-attention and external-attention intrinsic neural timescales; advanced meditators showed a smaller task difference, and that reduction correlated with reported nondual experience [69]. These studies show that parts of the phenomenology can be measured. They do not establish that meditation and seizure-coma states share one mechanism, that content-free awareness is literally contentless, or that either result validates a metaphysical interpretation.

The exact phrase “a color that was no color” has not been recovered in the primary hospital-account sources reviewed here. The recovered scene is explicitly blue and pink. If a separate primary source for the colorless-field report is found, the SAN prediction should be stated operationally: a formless or uniform field should involve reduced category-differentiating structure relative to ordinary color perception, not merely lower total activity.

9.8.2 Altered-state experiment

The decisive design crosses source constraint with phase organization while matching mean rate and power as closely as possible. Conditions should include externally constrained perception, sensory-deafferented or imagery-driven content, task-appropriate within-assembly alignment plus structured between-assembly separation, phase-scrambled dispersion, and a clinically observed excessive-synchrony condition where intervention is not experimentally induced. Endpoints must remain separate: content identity, vividness, confidence, internal-versus-external source judgment, body ownership, self-location, autobiographical self-access, report, and action.

The CDSH prediction is not that one condition maximizes every endpoint. It is that the structured receiver-relative model explains which distinctions survive, which merge, and which gain causal access better than global coherence, entropy, or frequency-band labels. The hypothesis fails if equal-capacity recurrent and state-space baselines capture the endpoints without the declared coherence-dispersion variables, if structured separation cannot be distinguished from noise, or if perturbation effects reduce to nonspecific arousal, movement, medication, source, or power changes.

9.9 Rival explanations and evidence ceilings

Domain	Established or rival result	What it can explain without CDSH	What would discriminate CDSH
Color	Early visual areas can encode stimulus chromaticity, while V4/VO1 activity can track perceived color under fixed retinal input [29,40].	Distributed stimulus and percept decoding.	Phase-boundary variables must predict and causally alter hue identity beyond activation, adaptation, attention, and report.
Umami	T1R1/T1R3 contributes to umami transduction, but residual responses after T1R3 deletion imply additional receptors or pathways [30,41].	Parallel receptor and pathway coding.	The proposed receiver-relative timing structure must add value beyond receptor identity, concentration, firing, and behavior.
Whisking	Touch count and whisking midpoint can explain fine object localization [42].	Learned active sensing with low-dimensional sensorimotor variables.	CDSH measures must improve held-out prediction or perturbation effects beyond that model.
Artificial touch	ICMS can evoke body-localized percepts, and biomimetic patterns can increase naturalness, but effects vary by electrode group and were tested in three participants [26-28,43].	Anatomical targeting and spatiotemporal stimulation design.	Receiver-state and coherence-dispersion variables must provide preregistered incremental causal value.
Report	P3b can disappear in a no-report condition despite visible stimuli [45].	Postperceptual decision and report processes.	Candidate content states must generalize across report and no-report conditions and predict nonverbal consequences.
Cross-person alignment	Hyperalignment can map idiosyncratic cortical topographies into a shared representational space [44].	Between-person decoding of common stimulus structure.	Translation must survive common-space and same-behavior nulls and reproduce recipient-specific causal consequences; shared decoding alone is insufficient.

These results narrow the claim rather than erase it. Altered states add a second constraint: global coherence, one frequency band, or one entropy statistic cannot stand in for the typed content, source, self-location, access, report, and action variables. CDSH earns support only from incremental, held-out, and preferably causal performance over the strongest domain-specific alternative.

10. Multiscale Predictions

10.1 Micro scale

1. The same presynaptic event will produce different postsynaptic and plasticity consequences depending on receiver phase, membrane state, dendritic location, and recent timing history.
2. Coherence and dispersion will alter the distribution of pre-post timing events rather than map universally to LTP and LTD.
3. Unexpected inhibition and omitted expected activity will sometimes predict learning or behavioral change as well as unexpected excitation.

10.2 Meso scale

1. Task-relevant local assemblies will show higher within-assembly phase consistency than matched nonparticipating cells.
2. Boundaries between competing assemblies will show reproducible phase offsets, inhibitory influence, or short desynchronization structure rather than arbitrary noise.
3. Mean synchrony will underperform a model that jointly measures within-assembly coordination and between-assembly structured dispersion.

10.3 Macro scale

1. Perceptual or decision states will correspond to transient, distributed coherence-dispersion topologies rather than one globally coherent rhythm.
2. Similar spectral power can support different contents when the topology of relative phase and effective connectivity differs.
3. Action and reafferent feedback will update the topology in a content- and outcome-dependent sequence.

10.4 Plasticity

Repeatedly imposing a specific within-assembly / between-assembly timing relation should bias later effective connectivity or response readiness, but the direction should follow measured local plasticity rules rather than a universal coherence / LTP assumption.

11. Experimental Program

Experiment 1: Phase-sensitive plasticity replication and extension

Preparation: cortical or hippocampal slice with identified pathways and cell classes.

Manipulation: deliver matched presynaptic activity at controlled postsynaptic oscillatory phases; add stable phase-offset and phase-scrambled population conditions.

Controls: identical spike counts, delivered charge, mean frequency, membrane potential, and neuromodulatory state.

Measures: immediate EPSP change, 1-hour change, multiday persistence where technically possible, dendritic calcium, and inhibitory recruitment.

CDSH test: structured phase relations should explain plasticity beyond rate and average synchrony, with pathway-specific rather than universal signs.

Experiment 2: Assembly and boundary perturbation

Preparation: animal sensory cortex with high-density electrophysiology and cell-type-specific perturbation.

Task: discriminate stimuli designed to recruit overlapping assemblies.

Manipulation: selectively increase within-assembly alignment, impose controlled between-assembly phase offsets, disrupt inhibitory boundaries, or phase-scramble while preserving rate and power.

Measures: decoding, discrimination, reaction time, assembly membership, effective connectivity, and later response changes.

CDSH test: best performance should occur at a task-specific organization of local coherence and structured separation, not necessarily maximal global synchrony.

Experiment 3: Human intracranial observational test

Participants: patients already undergoing clinically indicated intracranial monitoring.

Task: repeated perception, imagery, memory, and choice conditions with matched sensory energy.

Analysis: compare conventional power, coherence, and connectivity models with a preregistered topology model containing within-cluster phase consistency, stable between-cluster offsets, transient desynchronization duration, and directed influence.

CDSH test: the topology model must improve held-out content and behavior prediction beyond conventional variables. This observational experiment cannot alone establish causality or qualia.

Experiment 4: Artificial wrist-tap intervention

Goal: evoke and verify a localized wrist-tap percept through one medically appropriate intervention route.

Candidate routes: peripheral nerve, contralateral ventral posterolateral thalamus, or somatotopically appropriate S1, tested as alternatives rather than one serial artificial pathway.

Variables: site, amplitude or charge, pulse width, frequency, sequence, prestimulus tonic state, within-population alignment, and structured phase dispersion.

Evidence ladder: report -> localized forced choice -> quality / intensity discrimination -> behavioral use -> reversibility and safety.

CDSH test: receiver-state and phase-organization variables must add held-out predictive or causal value beyond anatomy and waveform alone. Existing ICMS evidence shows that artificial touch is often not experienced as natural and that biomimetic spatiotemporal stimulation can improve naturalness in a small participant sample [43]; this is a constraint and precedent, not validation of CDSH.

Experiment 5: Redness identity versus intensity

Goal: separate color identity from intensity, salience, and report preparation.

Task: present equiluminant hues and independently vary contrast, adaptation, attention, and report requirements.

Measures: population decoding, phase topology, recurrent directed influence, forced choice, priming, autonomic response, and confidence.

Perturbation: apply temporally targeted, medically and ethically appropriate disruption to a preregistered content-bearing route while preserving stimulus energy.

CDSH test: hue identity should correspond to a reproducible distributed topology that generalizes across intensity and report conditions better than power or mean synchrony alone. Perturbing a predicted relation should selectively alter hue discrimination or generalization.

Experiment 6: The broccoli reconstruction ladder

Goal: test the multimodal construction claim without assuming whole-object transfer.

Natural conditions: vision-only, odor-only, taste-only, texture-only, congruent broccoli, and incongruent property combinations.

Model: estimate subject-specific transformations and cross-modal recurrence using held-out sessions and active sampling behavior.

Intervention: begin with one low-dimensional, clinically justified property such as localized texture or taste-quality discrimination; do not proceed directly to a full object claim.

Measures: property-specific forced choice, cross-modal priming, object recognition, motor use, confidence, persistence, and adverse effects.

CDSH test: combining causally validated components should produce predictable integration or conflict

effects. A model that only decodes labels but cannot predict perturbation and recombination fails the stronger reconstruction claim.

Experiment 7: Cross-person translation under strong null models

Goal: test whether a learned transformation between two people predicts and causally reproduces a bounded sensory discrimination without treating common decoding as shared qualia.

Training: estimate subject-specific encoders and a transformation $T_{i \rightarrow j}$ from person i 's state to person j 's state using separate sessions and object families.

Nulls: anatomical coordinate transfer, marginal-distribution matching, same-report or same-behavior matching, label shuffling, and a hyperalignment / common-space model [44].

Held-out test: use new stimulus exemplars and task contexts, then compare the learned transformation with the strongest null N_{best} :

$$\Delta_{\text{trans}} = \mathcal{L}_j(N_{\text{best}}(Z_i), Y_j) - \mathcal{L}_j(T_{i \rightarrow j}(Z_i), Y_j). \quad (19)$$

Positive Δ_{trans} means that the learned mapping improves recipient-level prediction. It does not show that two people have identical qualia. The stronger causal test requires a medically justified intervention that reproduces the recipient's property-specific discrimination, generalization, action bias, and reafferent update while preserving safety and defeating the null models.

12. Analysis Requirements

1. Avoid treating volume-conducted zero-lag coherence as evidence of direct communication.
2. Use source separation, bipolar or local referencing where appropriate, and sensitivity analyses across referencing schemes.
3. Separate phase consistency from power and waveform shape.
4. Distinguish stable splay organization from random phase dispersion.
5. Control firing rate, arousal, movement, sensory input, and task difficulty.
6. Use held-out prediction and preregistered model comparisons.
7. Report negative and region-specific results rather than averaging them into a global claim.
8. Match the interpretation to the measurement scale; scalp EEG cannot establish single-synapse timing rules.
9. Separate perceptual content from report preparation with no-report, delayed-report, and nonverbal readouts where feasible [45].
10. For cross-person models, compare against hyperalignment or another common representational-space baseline and test new stimulus families [44].
11. Treat decoder accuracy as correlation unless a selective perturbation changes the preregistered recipient-level consequence.

13. Falsification Criteria

CDSH is weakened or falsified in a declared domain if:

1. anatomy, firing rate, spectral power, and ordinary state variables explain outcomes without phase-organization measures;
2. structured dispersion cannot be distinguished from random noise or adds no reproducible information;

3. perturbing within-assembly coherence or between-assembly separation does not alter routing, behavior, or later response readiness;
4. purported phase effects disappear after controlling waveform shape, power, reference choice, or common input;
5. repeated timing organization does not predict any later plasticity under conditions where plasticity is otherwise measurable;
6. effects fail to replicate across participants, preparations, or measurement methods; or
7. neural topology predicts task variables but not qualitative content better than conventional alternatives;
8. candidate observer-action states fail report-independent generalization or selective causal intervention; or
9. cross-person mappings do not outperform anatomical, behavioral, and common-space null models on held-out stimulus families.

Failure of the strongest phenomenal claim would not invalidate established phase-sensitive plasticity or communication-through-coherence. The proposal is deliberately decomposed so component evidence cannot be misrepresented as confirmation of the joined theory.

14. Medical and Conceptual Boundaries

- The paper is a mechanistic hypothesis, not clinical guidance.
- Coherence is not synonymous with health, accuracy, awareness, or cognition; excessive synchrony can be pathological.
- Desynchronization is not synonymous with noise or failure.
- LTP is not simply memory storage, and LTD is not simply forgetting or inhibition.
- A low population order parameter does not prove a splay state.
- EEG frequency bands do not map one-to-one onto thoughts or anatomical layers.
- Feynman’s path integral is not evidence for quantum consciousness.
- Friston’s framework does not prove SAN’s proposed oscillatory implementation.
- Distributed observer-action does not require an internal screen or localized viewer.
- Color decoding does not establish a sufficient neural mechanism for redness.
- Sensory substitution does not establish transfer of a pre-existing qualia code.
- Receptor cascades constrain SAN’s molecular account but do not prove its network or phenomenal claims.
- Neural stimulation is an invasive medical intervention and must remain under independent clinical, ethical, and regulatory governance.
- A shared representational space or successful between-person decoder does not establish identical experience.
- A report-related neural signature can reflect decision, memory, or motor preparation rather than perceptual awareness [45].
- Passing the proposed observer-action gate is evidence of causal participation in a tested loop, not a general consciousness detector.

15. Discussion

The central change introduced by CDSH is to make boundaries computationally meaningful. A theory based only on coherence can explain how selected populations communicate but risks treating everything outside the selected coalition as failed communication. SAN instead proposes that stable separation is part of the representation. The currently experienced state is sculpted not only by what participates together but by what remains phase-separated, inhibited, delayed, or omitted.

This creates a natural bridge between fast neural organization and slower learning. Fast coherence-dispersion patterns determine which signals arrive together, which are excluded, and which produce receiver-relative surprise. Those timing histories interact with local plasticity rules. Changed connectivity then alters the assemblies and boundaries available in the next cycle. The result is an iterative, embodied process: perception changes action; action changes sensory input; repeated differences change the network that will perceive later differences.

The proposal also clarifies SAN's tonic-canvas and phasic-ink language. The tonic canvas is not a globally synchronized carrier devoid of content. It is the current organized field of expected relations. The phasic ink is any consequential difference from that field, including excitation, inhibition, phase shift, changed coupling, or boundary crossing. Qualitative differentiation can therefore arise from both reinforcement and separation without invoking a homunculus.

The Feynman and Friston comparisons should remain disciplined. Feynman's formalism shows that phase relations can determine reinforcement and cancellation across possible histories. Friston supplies a formal account of prediction, action, and mismatch reduction. SAN proposes that receiver-relative phase differences, coherence-dispersion topology, and plasticity provide a biological implementation connecting these intuitions. That joining is the hypothesis to test, not a conclusion licensed by either donor theory.

16. Conclusion

The Coherence-Dispersion Sculpting Hypothesis proposes that neural computation and qualitative differentiation depend on both together-firing and persistently separated phase relations. Local coherence can recruit and bind; structured dispersion and inhibition can separate, suppress, and redirect. Phase-wave differentials alter those relations against an ongoing tonic context. Repeated timing histories can bias plasticity, while plasticity reshapes the future organization of coherence and dispersion.

The hypothesis is more specific than the claim that synchrony supports consciousness. It predicts that content and behavior depend on a reproducible topology of coherent assemblies and structured boundaries; that this topology outperforms global synchrony measures; and that imposing selected relations changes later routing in accordance with local plasticity rules. These predictions are experimentally accessible and falsifiable. Their evaluation can determine whether coherence-dispersion sculpting is merely a useful metaphor, a general neural coordination principle, or part of the mechanism by which a distributed biological network constructs differentiated experience and action.

Figure Plan

Figure 1. The bidirectional operator

Two-cell and multi-cell panels showing together-firing coherence, stable phase-offset coupling, structured play, and random phase noise. The figure must distinguish low global order from lack of organization.

Figure 2. Micro-meso-macro scale separation

Synaptic timing and dendritic state at micro scale; coherent assemblies and inhibitory / phase boundaries at meso scale; distributed sensory-association-body-action topology at macro scale. No universal cortical column and no inner viewer.

Figure 3. Fast organization and slow plasticity loop

Current tonic state -> PWD event -> recruitment / separation -> action and refference -> timing-dependent plasticity -> changed future routing.

Figure 4. Feynman, Friston, and SAN boundary diagram

Three columns: established path-integral phase interference; free-energy / predictive-processing framework; proposed SAN neural implementation. Arrows must be labeled analogy, conceptual bridge, and testable hypothesis rather than proof.

Figure 5. Decisive assembly-boundary experiment

Matched-rate and matched-power conditions comparing increased within-assembly alignment, structured between-assembly dispersion, and phase-scrambled control, with behavioral and plasticity outcomes.

Figure 6. Artificial wrist-tap test

Natural tactile reference, separate candidate intervention routes, controlled waveform and receiver-state variables, and evidence ladder ending in distributed behavioral use.

Figure 7. Cross-person translation and null-model ladder

Two subject-specific representational spaces, a learned transformation, anatomical and common-space baselines, held-out object families, and the separate prediction, intervention, and phenomenal-identity claim levels.

Figure 8. Altered-state dynamic range

A two-dimensional map separates within-assembly coordination from structured between-assembly separation and places ordinary perception, internally generated content, excessive synchrony, and unstructured dispersion as test conditions rather than diagnoses. Side panels keep content, source attribution, bodily self-location, autobiographical access, report, and action as separate endpoints.

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SAN Primary-Source Appendix

The following local files are source-genealogy evidence, not independent scientific validation:

- a0041z.md - first Git commit 7d1431213cd4ac9788848a5396ec2992ece11cfd, June 8, 2022; current-file SHA-256 45F120F288BD7C3FFCE86830301CC35EB2B988F12AE44A62B1B22BE0A156E01C.
- a0142z.md - Git commit ff8d45af7706a29baaef326fcc4fbcbaa0ed0f06, June 11, 2022; current-file SHA-256 601BA2C6812BB00EC460F98F2F45274FFEC789F100F67BB62CF4C0DFBC371A0C.
- a0601z.md - Git commit 5360f6414909a2474129b2140ee0dc3112c93b6b, June 8, 2022; current-file SHA-256 622CC88DC8218ED784FC69043446FF5DB15F7EBBCDC0AF51AEB1F1863B449C65.
- 02san.md - later joined PWD account; current-file SHA-256 ADD0F593BC0BA289907749D2D290A59556410046D7203A2F94DFEB2CDC6D8706.
- raynote16.md - first Git commit 049381ed44008f3e32a259c4aaea6b3b389b60f9, January 3, 2025; current-file SHA-256 F9BEB7622ECBC75E45289B98EDFE996478E90A7920B388FF1FBD9441A9B053D4.

Current-file SHA-256 values were recorded during drafting so later edits can be distinguished from the source snapshot used here. Proposition-level dates and claim ceilings are governed by reviews/DRAFT4-IMMUTABLE-SOURCE-ATOM-AUDIT-20260720.md.

The Qualia construction chain additionally depends on these dated sources:

- n1p1.md - the April 2017 TCP / UDP protocol question and deliberate ATP / Alternating Transmission Protocol double meaning at lines 277-451; original public audio route: https://www.youtube.com/watch?v=YrX_68oKuVs. Local content is recovered; immutable public-date sealing remains open.
- a0037z.md - relative tonic / phasic roles; commit 226714e92e968718db74ea6299268e357952d21d, June 10, 2022.
- a0306z.md - the early broccoli / BCI route is present in commit 81210149c4fbf14575ee6c6b56c05daa34a92273 on June 12, 2022. Explicit redness / tone, integrated broccoli, reconstruction, and mature NAPOT claims are separately anchored to August 23 commits 35f21624, 368b0a6d, fc922f14, 24761c80, f9ce9181, db79914b, f6602d79, 5caffef52, 57d5bfeb, 48649b10, and e076d8aa.
- b0100y.md - distributed observation, capture / projection analogies, and rejection of a separately rendered observer; key proposition commits are d176623b, 0ba414d0, and 41562c5f, June 18-19, 2022.
- a0011z.md - mechanosensory receptor / threshold ancestry is fixed by 9533a084 and 6cc9c674 on June 29, 2022 and expanded by 9911b61b on August 22. The exact current action-potential-duration sentence is commit bf6e02b5, January 4, 2024.
- a0272z.md - umami and receptor-diversity bridge; commit 75b194fb97d2da09fb8bdd4130be98b910a72adb, August 10, 2022.
- b0281y.md - olfactory tempo-spatial properties are fixed June 17, 2022 in 61d9e96d and 38776366; the explicit phase-field / NAPOT synthesis is commit 219bdf24, June 18, 2022.
- a0296zWhisker.md - whisker evidence intake begins in commit 29369061baee78f8974e30a8e48f1bc95ae91fa2, September 4, 2022; the explicit SAN desynchronization / Qualia interpretation is commit 55cb3bcb0ea8653087b424444e2f1cf66ca1d580, November 29, 2022.

- a0308z.md - redness, real and imagined content, and temporary distributed rendering; Git-added November 28, 2022.
- a0325zNAPOT6.md - ions-to-qualia, sensory / motor qualia, persistence, and magnitude; Git-added November 28, 2022.

Authorship and Historical Provenance

The core ideas, scientific theories, hypotheses, and original source program developed in this paper are Micah Blumberg's work. The research spans 2011 through 2026, and the central SAN Qualia architecture documented here was developed and publicly recorded primarily before ChatGPT's public release in November 2022. AI systems did not originate that research program. They have been used under the author's direction for limited assistance with source recovery, mathematical and coding formulations, grammar and spelling, and checks against medical, biological, and physical literature. Responsibility for the claims, interpretations, and final manuscript remains with the author.